Fundamental of Mathematics

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Inequalities and Modulus Function

Let $y = \frac{f(x)}{g(x)}$ be an expression in x where f(x) and g(x) are polynomials in x. If it is given that $y > 0$ (or $y < 0$) then we have to write set of all the values of x for which y is positive (or y is negative) to evaluate this set, we follow these steps.

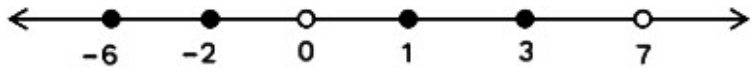
Step-1 :

Express f(x) and g(x) as factor of linear expressions (If possible) also collect the same linear factors in one place.

Ex. Let after step-1 $y = \frac{f(x)}{g(x)} = \frac{(x-1)^3 (x+2)^4 (x-3)^5 (x+6)}{x^2 (x-7)^3}$

Step-2 :

Now mark the values of x for which y = 0 by black dots on number line (x = 1, –2, 3, –6). Also mark the values for which y is undefined by white dots (x = 0, 7)



Step-3 :

Now check value of y for any real number greater than the right most marked number on number line. If it is positive, then $y > 0$ for all numbers greater than the right most marked number.

Ex. Here if we put $x = 8$ in y then $y > 0$ hence $y > 0 \quad \forall\, x \in (7, \infty)$

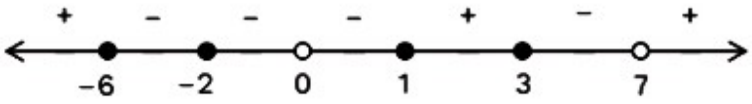
Step-4 :

If exponent of a factor in y is odd, then this point is called simple point and if the exponent is even, then the point is called double point. For above example simple points are $x = 1, 3, -6, 7$

double points are $x = 0, -2$

Now sign in neighbouring interval of simple point is different and for double point it is same.

For above example sign scheme will be following



Step-5 :

$y > 0$ for all values of x which lies in interval where positive mark is present and $y < 0$ for all values of x of interval where negative mark is present. Hence solution of

$y > 0$ is $x \in (-\infty, -6) \cup (1, 3) \cup (7, \infty)$
$y \geq 0$ is $x \in (-\infty, -6] \cup \{-2\} \cup [1, 3] \cup (7, \infty)$
$y < 0$ is $x \in (-6, -2) \cup (-2, 0) \cup (0, 1) \cup (3, 7)$
$y \leq 0$ is $x \in [-6, 0) \cup (0, 1] \cup [3, 7)$

Point to Remember!!!

⇒ Points where denominator is zero are never included in the answer.
⇒ Never cross multiply in inequality until you are sure that denominator will always be positive.
⇒ We can also multiply both side by negative expression by changing the sign of inequality.
⇒ We can add and subtract the same expression on both side without changing the sign of inequality.

Q. Solve the following inequalities for $y > 0$, $y \geq 0$, $y < 0$ and $y \leq 0$.

(i) $y = \dfrac{(x+1)^2(2x-3)}{(x+7)^3}$

(ii) $y = \dfrac{(x^2-5x+6)(x^2-8x+15)}{(x^2-16)}$

(iii) $y = \dfrac{(3x-2)(4-x)^3}{(7-2x)^2(5-x)^2}$

(iv) $y = \dfrac{(x^3-6x^2+11x-6)}{(x^2+2)(x^2-8x+12)}$

(v) $y = \dfrac{(x^2-3)(x^4+x^2+1)(2^x-1)}{(3x+1)^7(x+5)^2}$

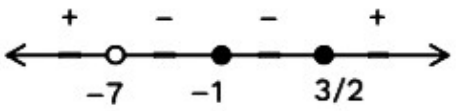
(vi) $y = (x^2-6x-2)$

Sol. **(i)** $y = 0$ for $x = -1, \dfrac{3}{2}$ (black dots)

y not defined for $x = -7$ (white dot)

Single points $x = \dfrac{3}{2}, -7$

Double points $x = -1$



$y > 0 \Rightarrow x \in (-\infty, -7) \cup \left(\dfrac{3}{2}, \infty\right)$

$y \geq 0 \Rightarrow x \in (-\infty, 7) \cup \{-1\} \cup \left[\dfrac{3}{2}, \infty\right)$

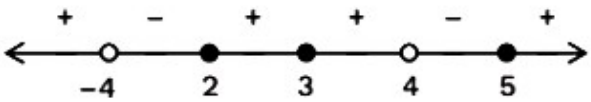
$y < 0 \Rightarrow x \in (-7, -1) \cup \left(-1, \dfrac{3}{2}\right)$

$y \leq 0 \Rightarrow x \in \left(-7, \dfrac{3}{2}\right]$

(ii) $(x^2 - 5x + 6) = (x - 2)(x - 3)$

$(x^2 - 8x + 15) = (x - 5)(x - 3)$

$(x^2 - 16) = (x + 4)(x - 4)$



$$y = \frac{(x-2)(x-3)^2(x-5)}{(x+4)(x-4)}$$

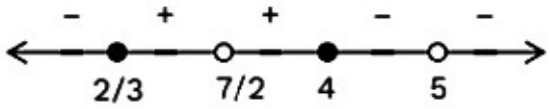
$y > 0 \Rightarrow x \in (-\infty, -4) \cup (2, 3) \cup (3, 4) \cup (5, \infty)$

$y \geq 0 \Rightarrow x \in (-\infty, -4) \cup [2, 4) \cup [5, \infty)$

$y < 0 \Rightarrow x \in (-4, 2) \cup (4, 5)$

$y \leq 0 \Rightarrow x \in (-4, 2] \cup (4, 5]$

(iii)



(Note if we put any number $x > 5$ in y, then $y < 0$.
Hence right-most region is negative)

$$y > 0 \Rightarrow x \in \left(\frac{2}{3}, 4\right) - \left\{\frac{7}{2}\right\}$$

$$y \geq 0 \Rightarrow x \in \left[\frac{2}{3}, 4\right] - \left\{\frac{7}{2}\right\}$$

$$y < 0 \Rightarrow x \in \left(-\infty, \frac{2}{3}\right) \cup (4, \infty) - \{5\}$$

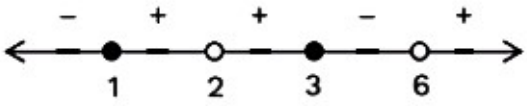
$$y \leq 0 \Rightarrow x \in \left(-\infty, \frac{2}{3}\right] \cup [4, \infty) - \{5\}$$

(iv) $x^3 - 6x^2 + 11x - 6 = (x-1)(x-2)(x-3)$

$(x^2 + 2) \rightarrow$ Always positive

$$x^2 - 8x + 12 = (x-6)(x-2)$$

$$y = \frac{(x-1)(x-3)}{(x^2+2)(x-6)(x-2)^\circ}$$



$$y > 0 \Rightarrow x \in (1,3) \cup (6, \infty) - \{2\}$$

$$y \geq 0 \Rightarrow x \in [1,3] \cup (6, \infty) - \{2\}$$

$$y < 0 \Rightarrow x \in (-\infty, 1) \cup (3, 6)$$

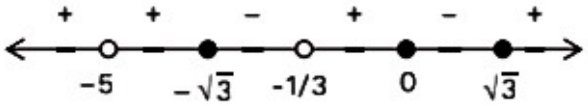
$$y \leq 0 \Rightarrow x \in (-\infty, 1] \cup [3, 6)$$

(v) $x^4 + x^2 + 1 \Rightarrow$ always positive

$2^x - 1$ can be replaced by x because both have same sign scheme on number line

$$x^2 - 3 = \left(x + \sqrt{3}\right)\left(x - \sqrt{3}\right)$$

$$y = \frac{\left(x+\sqrt{3}\right)\left(x+-\sqrt{3}\right)\left(x^4+x^2+1\right)x}{(3x+1)^7 (x+5)^2}$$



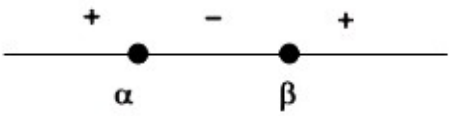
$$y > 0 \Rightarrow x \in \left(-\infty, -\sqrt{3}\right) \cup \left(-\frac{1}{3}, 0\right) \cup \left(\sqrt{3}, \infty\right) - \{-5\}$$

$$y \geq 0 \Rightarrow x \in \left(-\infty, -\sqrt{3}\right] \cup \left(-\frac{1}{3}, 0\right] \cup \left[\sqrt{3}, \infty\right) - \{-5\}$$

$$y < 0 \Rightarrow x \in \left(-\sqrt{3}, -\frac{1}{3}\right) \cup \left(0, \sqrt{3}\right)$$

$$y \leq 0 \Rightarrow x \in \left[-\sqrt{3}, -\frac{1}{3}\right) \cup \left[0, \sqrt{3}\right]$$

(vi) Let $\left(x^2 - 6x - 2\right) = (x - \alpha)(x - \beta)$



$$\alpha, \beta = \frac{6 \pm \sqrt{36 - 4(-2)}}{2} \Rightarrow \alpha, \beta = \frac{6 \pm 2\sqrt{11}}{2}$$

$$\alpha = 3 - \sqrt{11}, \quad \beta = 3 + \sqrt{11} \quad (\alpha < \beta)$$

$$y > 0 \Rightarrow x \in (-\infty, \alpha) \cup (\beta, \infty) = \left(-\infty, 3 - \sqrt{11}\right) \cup \left(3 + \sqrt{11}, \infty\right)$$

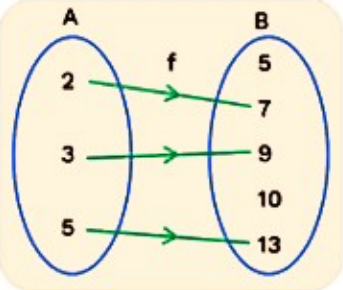
$$y \geq 0 \Rightarrow x \in (-\infty, \alpha] \cup [\beta, \infty) = \left(-\infty, 3 - \sqrt{11}\right] \cup \left[3 + \sqrt{11}, \infty\right)$$

$$y < 0 \Rightarrow x \in (\alpha, \beta) = \left(3 - \sqrt{11}, 3 + \sqrt{11}\right)$$

$$y \leq 0 \Rightarrow x \in [\alpha, \beta] = \left[3 - \sqrt{11}, 3 + \sqrt{11}\right]$$

FUNCTION :

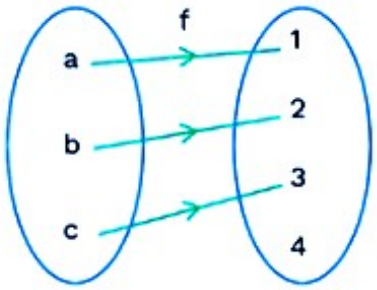
It is a relation from non-empty set A to non-empty set B such that each element of set A is uniquely associated with elements of set B.



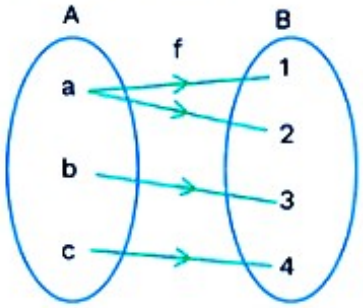
Ex.: f(x) = 2x + 3 is a function from set A = {2, 3, 5} to set B = {5, 7, 9, 10, 13}
Domain of f = A = {2, 3, 5}
Range of f = {7, 9, 13}
Range $\subseteq$ co-domain
"7" is called "Image" of "2" under 'f'
'2' is called "Pre-image" / "Argument" of '7' under 'f' and function f is denoted by
f : A → B

Q.1 Identify which mapping(s) is/are a function (f) from A to B

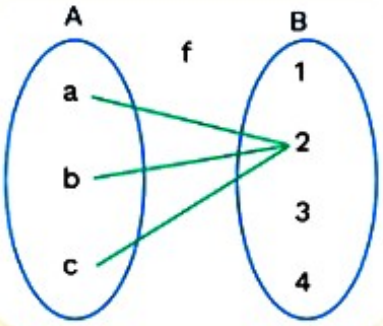
(i)



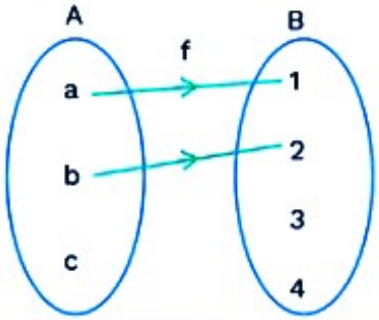
(ii)



(iii)



(iv)



Sol. **(i)** It is a function
(ii) It is not a function, ∵ a has 2 images
(iii) It is a function
(iv) It is not a function, ∵ $f(c) \in \phi$

Q.2 Find domain of the following functions:

(i) $f(x) = \sqrt{x}$ **(ii)** $f(x) = \sqrt{x^2}$ **(iii)** $f(x) = \sqrt{x^2 + 1}$

(iv) $f(x) = \sqrt{2x - 1}$ **(v)** $f(x) = \dfrac{1}{3x + 1}$

Sol.
(i) $x \geq 0$ $\Rightarrow D_f \in [0, \infty)$
(ii) $x^2 \geq 0$ $\Rightarrow D_f \in R$
(iii) $x^2 + 1 \geq 0$ $\Rightarrow D_f \in R$
(iv) $2x - 1 \geq 0$ $\Rightarrow D_f \in [½, \infty)$
(v) $3x + 1 \neq 0$ $\Rightarrow D_f \in R - \{-1/3\}$

Q.3 Find range of following functions:

(i) $f(x) = \sqrt{x^2 + 1}$ **(ii)** $f(x) = \sqrt{x^2 - 1}$ **(iii)** $f(x) = \sqrt{x^2 + 4} + 2$

Sol.

(i) $\because x^2 \geq 0$
$\therefore x^2 + 1 \geq 1 \quad \Rightarrow \sqrt{x^2 + 1} \geq 1$
$\therefore$ Range $\in [1, \infty)$

(ii) $\because x^2 \geq 0$
$\therefore x^2 - 1 \geq -1 \quad \Rightarrow \sqrt{x^2 - 1} \geq 0$
$\therefore$ Range $\in [0, \infty)$

(iii) $\because x^2 \geq 0$

$\therefore x^2 + 4 \geq 4 \quad \Rightarrow \sqrt{x^2+4} \geq 2$

$\sqrt{x^2+4} + 2 \geq 4$

$\therefore$ Range $\in [4, \infty)$

MODULUS FUNCTION

It is also known as absolute value function and absolute value of a real number x is denoted by |x|.

Definition

$$|x| = \begin{cases} x & ; \ x \geq 0 \\ -x & ; \ x < 0 \end{cases}$$

Domain: $x \in R$, **Range: $|x| \in [0, \infty)$**

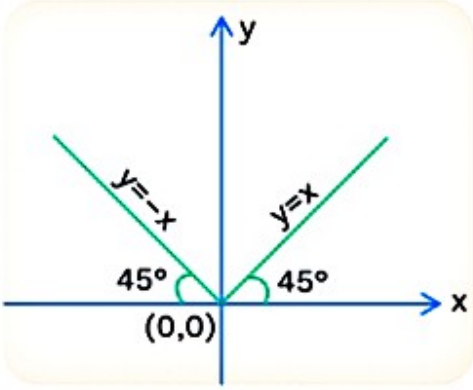
If $|f(x)| = \begin{cases} f(x) & ; \ f(x) \geq 0 \\ -f(x) & ; \ f(x) < 0 \end{cases}$

Note :

Values of x for which f(x) = 0 are critical points for |f(x)|

Graph:

Let $y = |x|$



Note :

Graph of y = |x| is symmetric about x = 0

Ex. $|2| = 2, \quad |5| = 5, \quad |\sqrt{3}| = \sqrt{3}$

$|-3| = -(-3) = 3, \quad |-4| = -(-4) = 4, \ldots$ etc

Q.1 Define the following functions:
(i) $|2x-1|$ **(ii)** $|x^2-5x-24|$

Sol. **(i)** $$|2x-1| = \begin{cases} 2x-1 & ; \quad 2x-1 > 0 \\ -(2x-1) & ; \quad 2x-1 < 0 \end{cases}$$

$$= \begin{cases} 2x-1 & ; \quad x > \frac{1}{2} \\ 1-2x & ; \quad x < \frac{1}{2} \end{cases}$$

(ii) $x^2-5x-24 = (x-8)(x+3)$



$$|x^2-5x-24| = \begin{cases} x^2-5x-24 & ; \quad x \in (-\infty,-3] \cup [8,\infty) \\ -(x^2-5x-24) & ; \quad x \in (-3,8) \end{cases}$$

Q.2 Define the following:
(i) $|x-2|+|x-5|$ **(ii)** $|x+1|-|2x-1|+|x-5|$

Sol. **(i)** Critical point for $|x-2|$ is $x=2$
Critical point for $|x-5|$ is $x=5$

$$|x-2|+|x-5| = \begin{cases} (x-2)+(x-5) & ; \quad x \geq 5 \\ (x-2)-(x-5) & ; \quad 2 \leq x < 5 \\ -(x-2)-(x-5) & ; \quad x < 2 \end{cases}$$

$$= \begin{cases} 2x-7 & ; \quad x \geq 5 \\ 3 & ; \quad 2 \leq x < 5 \\ 7-2x & ; \quad x < 2 \end{cases}$$

(ii) Critical point $\Rightarrow -1, \frac{1}{2}, 5$

$$|x+1|-|2x-1|+|x-5| = \begin{cases} (x+1)-(2x-1)+(x-5) & ; \quad x \geq 5 \\ (x+1)-(2x-1)-(x-5) & ; \quad \frac{1}{2} \leq x < 5 \\ (x+1)+(2x-1)-(x-5) & ; \quad -1 \leq x < \frac{1}{2} \\ -(x+1)+(2x-1)-(x-5) & ; \quad x < -1 \end{cases}$$

$$= \begin{cases} -3 & ; \quad x \geq 5 \\ 7-2x & ; \quad \frac{1}{2} \leq x < 5 \\ 2x+5 & ; \quad -1 \leq x < \frac{1}{2} \\ 3 & ; \quad x < -1 \end{cases}$$

Q.3 Draw graph of

(i) $f(x) = |x - 3|$ **(ii)** $f(x) = |x + 1| + |x - 5|$

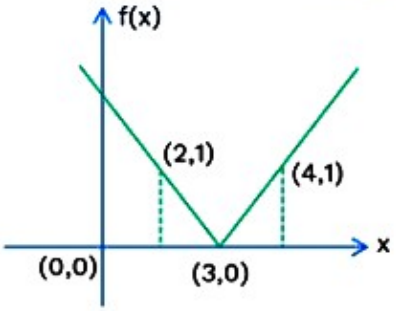
(iii) $f(x) = |x - 1| + |x - 3| + |x - 7|$

Sol. **(i)** Critical point $x = 3$

$f(3) = 0,$

$f(4) = |4 - 3| = 1,$

$f(2) = 2\,|2 - 3| = 1$



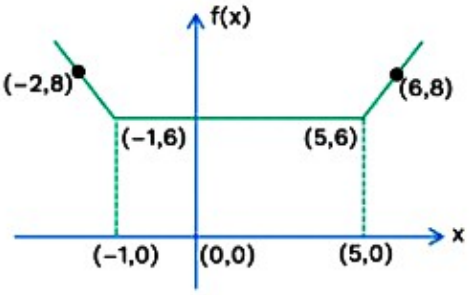
(ii) Critical points $x = -1, 5$

$f(-1) = 0 + 6 = 6,$

$f(5) = 6 + 0 = 6$

$f(6) = 7 + 1 = 8,$

$f(-2) = 1 + 7 = 8$



(iii) Critical points $x = 1, 3, 7$

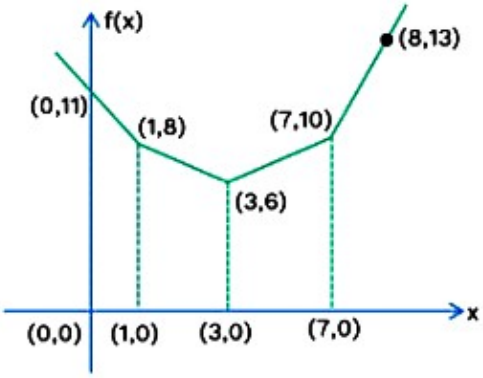
$f(1) = 8,$

$f(3) = 6,$

$f(7) = 10$

$f(0) = 11,$

$f(8) = 13$

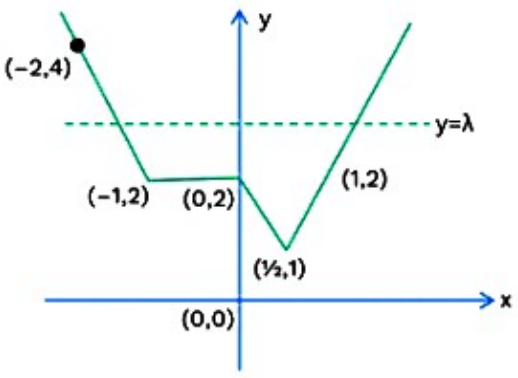


Q.4 Find λ if the equation $|x + 1| - |x| + |2x - 1| = \lambda$ have

(i) No solution **(ii)** Exactly one solution

(iii) Exactly two solutions **(iv)** More than two solution

Sol. To solve this problem, we will use graphical approach for $y = |x + 1| - |x| + |2x - 1|$ and $y = \lambda$



For **(i)** $\lambda < 1$ **(ii)** $\lambda = 1$

(iii) $\lambda \in (1, 2) \cup (2, \infty)$ **(iv)** $\lambda = 2$

Properties of modulus function :

1. $|a| = |-a|$

2. $|a - b| = |b - a|$

3. $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$

4. $|ab| = |a| \cdot |b|$

5. $|a^2| = |a|^2 = a^2$

6. $\sqrt{a^2} = |a|$

7. $|f(x)| = f(x) \Rightarrow f(x) \geq 0$

8. $|f(x)| = -f(x) \Rightarrow f(x) \leq 0$

9. $|a| = |b| \Rightarrow a = \pm b$

10. $|a + b| \leq |a| + |b|$
Equality holds when $ab \geq 0$

11. $|a - b| \geq ||a| - |b||$
Equality holds when $ab \leq 0$

12. $|x - a| < b$
$\Rightarrow (x - a < b) \cap (x - a > -b)$

13. $|x - a| > b$
$\Rightarrow (x - a > b) \cup (x - a < -b)$

14. $|a|$ represents distance between 0 and a on number line.

15. $|a - b|$ represents distance between a and b on number line.

Q.1 Solve the equations:

(i) $|x - 2| = 5$ **(ii)** $||x - 1| - 2| = 3$ **(iii)** $(x - 4)^2 + |x - 3| + 2 = 0$

(iv) $x^2 - 3|x| - 10 = 0$ **(v)** $|x^2 + 3x - 4| = |x^2 - 1|$ **(vi)** $|x^2 - 5x + 6| = 5x - x^2 - 6$

Sol.

(i) $x - 2 = 5$ or $x - 2 = -5$

$x = 7$ **or** **$x = -3$**

(ii) $|x - 1| - 2 = 3$ or $|x - 1| - 2 = -3$

$|x - 1| = 5$ or $|x - 1| = -1$

$x - 1 = 5, -5$ or Not possible

$x = 6, -4$

(iii) $(x - 4)^2 \geq 0, \quad |x - 3| \geq 0$

$\because$ LHS ≥ 2

$\therefore$ **No solution**

(iv) $|x|^2 - 3|x| - 10 = 0$

Let $|x| = t$

$t^2 - 3t - 10 = 0 \Rightarrow (t - 5)(t + 2) = 0$

$\Rightarrow t = 5, t = -2$

$\Rightarrow |x| = 5, \quad |x| = -2$ (Not possible)

$\Rightarrow$ **$x = 5, -5$**

(v) $x^2 + 3x - 4 = x^2 - 1$, $x^2 + 3x - 4 = -(x^2 - 1)$

$3x = 3$, $2x^2 + 3x - 5 = 0$

$x = 1$, $(2x + 5)(x - 1) = 0$

$x = -5/2, 1$

(vi) $|x^2 - 5x + 6| = -(x^2 - 5x + 6)$

$\Rightarrow x^2 - 5x + 6 \le 0$

$(x - 2)(x - 3) \le 0 \quad \Rightarrow x \in [2, 3]$

Q.2 Solve the equations:

(i) $|x| + |2x - 3| = 4$ **(ii)** $|2x + 1| + 3x = 11$

Sol. **(i)** **Case-I** $x \ge 3/2$

$x + 2x - 3 = 4 \Rightarrow x = \frac{7}{3}$

Case-II $0 \le x < 3/2$

$x - (2x - 3) = 4 \Rightarrow x = -1$ (Not possible)

Case-III $x < 0$

$-x - (2x - 3) = 4 \Rightarrow x = -\frac{1}{3}$

$\therefore x = -\frac{1}{3}, \frac{7}{3}$

(ii) $|2x + 1| = 11 - 3x$ ... (1)

$2x + 1 = 11 - 3x$, or $-(2x + 1) = 3x - 11$

$x = 2$ $x = 12$

(Not satisfying equation (1))

$\therefore x = 2$

Q.3 Solve the equation:

(i) $|x - 1| + |x + 1| = |2x|$ **(ii)** $||2x - 3| - |x^2 - 4x + 3|| = |x^2 - 2x|$

Sol. **(i)** $|x - 1| + |x + 1| = |(x - 1) + (x + 1)|$

$\because |a| + |b| = |a + b| \Rightarrow ab \ge 0$

$\therefore (x - 1)(x + 1) \ge 0$

$x \in (-\infty, -1] \cup [1, \infty)$

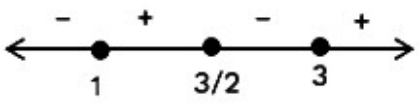
(ii) $||2x - 3| - |x^2 - 4x + 3|| = |(x^2 - 4x + 3) + (2x - 3)|$

$\because ||a| - |b|| = |a + b| \Rightarrow ab \le 0$

$\therefore (2x - 3)(x^2 - 4x + 3) \le 0$

$(2x - 3)(x - 1)(x - 3) \le 0$

$x \in (-\infty, 1] \cup [3/2, 3]$



Q.4 Solve the inequalities:
(i) $|x| < 3$ **(ii)** $|x - 2| > 5$ **(iii)** $|x + 5| < 0$ **(iv)** $|2x + 1| \leq 7$

Sol. **(i)** $-3 < x < 3$
$\Rightarrow x \in (-3, 3)$

(ii) $(x - 2) < -5$ and $(x - 2) > 5$
$x < -3$ and $x > 7$
$\Rightarrow x \in (-\infty, -3) \cup (7, \infty)$

(iii) $|x + 5|$ cannot be negative
$\Rightarrow x \in \phi$

(iv) $-7 \leq 2x + 1 \leq 7$
$-8 \leq 2x \leq 6$
$x \in [-4, 3]$

Greatest Integer Function (GIF)

Definition

It represents the greatest integer less than or equal to x and denoted by [x].

Ex. :

$[2] = 2$	$[5] = 5$	$[0] = 0$	$[2.1] = 2$
$[2.99] = 2$	$[-1] = -1$	$[-1.9] = -2$	$[-1.1] = -2$

$$[x] = \begin{cases} \vdots & \\ 2 & ; \quad 2 \leq x < 3 \\ 1 & ; \quad 1 \leq x < 2 \\ 0 & ; \quad 0 \leq x < 1 \\ -1 & ; \quad -1 \leq x < 0 \\ -2 & ; \quad -2 \leq x < -1 \\ \vdots & \end{cases}$$

Q.1 Evaluate the value of ([·] represents GIF)
(i) $[\pi] + [e]$
(ii) $[-e^2] + [\sin 91°]$
(iii) $\left[\frac{1}{3} + \frac{1}{90}\right] + \left[\frac{1}{3} + \frac{2}{90}\right] + \left[\frac{1}{3} + \frac{3}{90}\right] + \cdots + \left[\frac{1}{30} + \frac{90}{90}\right]$

Sol. **(i)** $\because \pi \approx 3.14$, $e \approx 2.71$
$[\pi] + [e] = 3 + 2 = \mathbf{5}$

(ii) $-e^2 \simeq -7.3$, $0 < \sin 91° < 1$
$[-e^2] + [\sin 91°] = -8 + 0 = \mathbf{-8}$

(iii) Let $f(r) = \left[\frac{1}{3} + \frac{r}{90}\right]$

$\because$ f(1) = f(2) = ... = f(59) = 0
f(60) = f(61) = ... = f(90) = 1

$\therefore$ Given expression becomes,

$$\underbrace{0+0+\ldots+0}_{59 \text{ terms}} + \underbrace{1+1+\ldots+1}_{31 \text{ terms}}$$

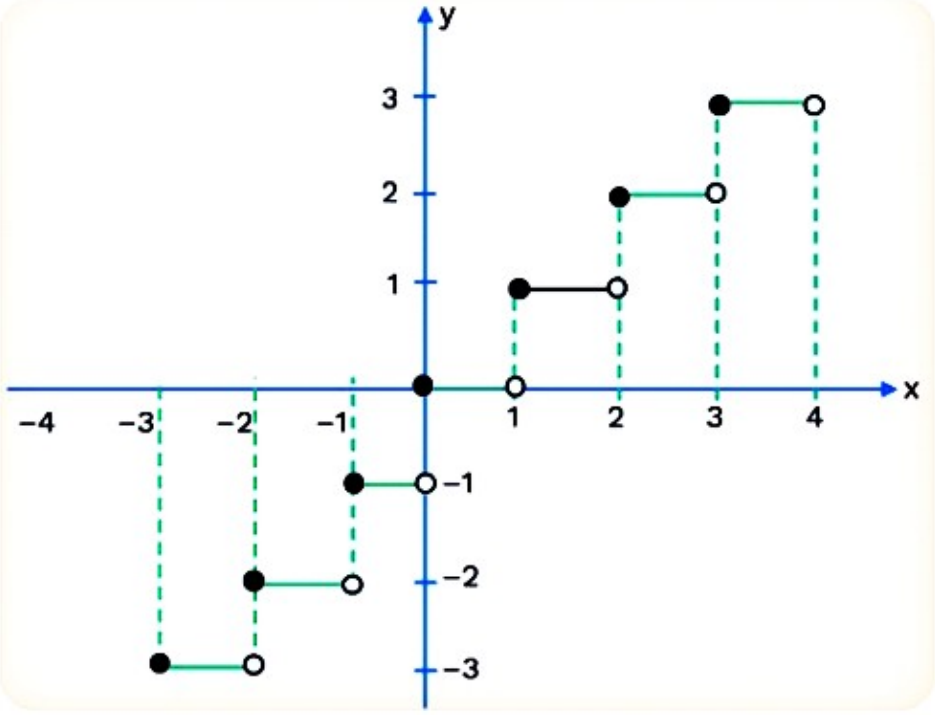
$= \mathbf{31}$

Graph:

Let y = [x] → GIF

Domain: x ∈ R

Range: [x] ∈ I



Properties: ([·] → GIF)

1. [x + m] = [x] + m (m ∈ I)

2. $[x]+[-x] = \begin{cases} 0 & ; \ x \in I \\ -1 & ; \ x \notin I \end{cases}$

Q.1 Solve the equations: ([·] → GIF)

(i) $[x] = 2$

(ii) $[x + 1] = 0$

(iii) $[x] + [-x] = -x^2$

Sol. **(i)** $[x] = 2$

$\Rightarrow \quad x \in [2, 3)$

(ii) $[x + 1] = 0$

$\Rightarrow \quad [x] + 1 = 0 \quad \Rightarrow \quad [x] = -1$

$\Rightarrow \quad x \in [-1, 0)$

(iii) Case-I If $x \in I$ then

$0 = -x^2 \quad \Rightarrow \quad x = 0$

Case-II If $x \notin I$ then

$-1 = -x^2 \quad \Rightarrow \quad x^2 = 1$

$\Rightarrow x = \pm 1$ (Not possible)

$\because x \notin I$

Hence $x = 0$ only

Fractional Part Function (FPF)

Definition

Fractional part of x denoted by {x} and it can be evaluated as $\{x\} = x - [x]$ (where $[\cdot] \to$ GIF)

Ex.:

$\{2\} = 2 - [2] = 2 - 2 = 0$

$\{1.7\} = (1.7) - [1.7] = (1.7) - 1 = 0.7$

$\{-1.7\} = (-1.7) - [-1.7] = (-1.7) - (-2) = 0.3$

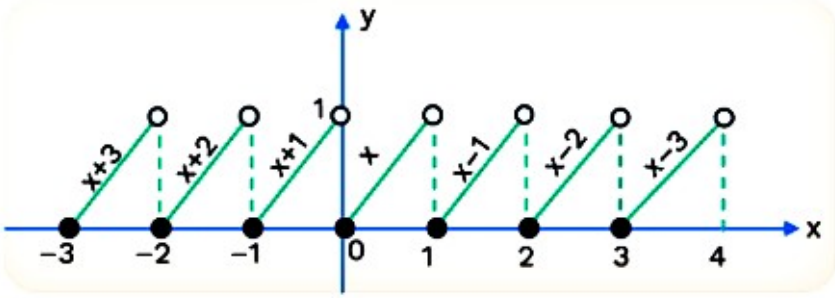
$$\{x\} = x - [x] = \begin{cases} x - 2 & ; \quad x \in [2, 3) \\ x - 1 & ; \quad x \in [1, 2) \\ x & ; \quad x \in [0, 1) \\ x + 1 & ; \quad x \in [-1, 0) \\ x + 2 & ; \quad x \in [-2, -1) \end{cases}$$

Graph :

Let $y = \{x\}$

Domain: $x \in R$

Range: $\{x\} \in [0, 1)$



Properties : ({·} → FPF, [·] → GIF)

1. $\{x + n\} = \{x\}, \quad n \in I$

2. $\{x\} + \{-x\} = \begin{cases} 0 & ; \ x \in I \\ 1 & ; \ x \notin I \end{cases}$

3. $\{[x]\} = [\{x\}] = 0$

4. Any real number $x = [x] + \{x\}$
or $x = I + f \quad (I = [x], f = \{x\})$

Q.1 Evaluate the value of ([·] → GIF, {·} → FPF)

(i) $\{\pi\}$

(ii) $\left\{\left[1+\frac{1}{2}\right]+\left[2+\frac{1}{3}\right]+\left[3+\frac{1}{4}\right]+\cdots+\left[100+\frac{1}{101}\right]\right\}$

Sol. **(i)** $\pi - [\pi] = \pi - 3$

(ii) {Integer} = 0

Q.2 Solve the equation ([·] → GIF, {·} → FPF)

(i) $\{x\} = ½$

(ii) $\{x + 2\} = 0$

(iii) $[x] + [-x] = \{x\} + \{-x\}$

Sol. **(i)** $x = [x] + \{x\} = \ldots, ½, 1 + ½, 2 + ½, \ldots$
$x = I + ½$

(ii) $\{x\} = 0$
$\Rightarrow x \in I$

(iii) **Case-I** if $x \in I$
$0 = 0$ true, hence $x \in I$

Case-II if $x \notin I$
$-1 = 1$ false $\Rightarrow x \in \phi$

hence, $x \in I$ is the solution.

Identity Function

Definition

It is a function which is defined by $f(x) = x \ \forall \ x \in R$

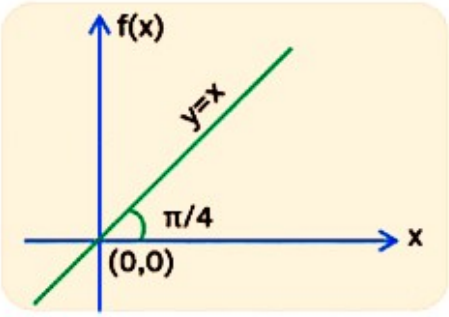
It is also known as identity relation or identity map or identity transformation.

Graph :

Let $f(x) = x$

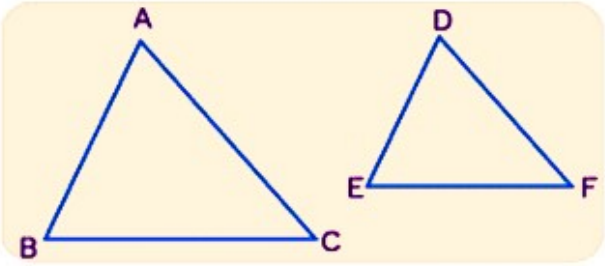
Domain: $x \in R$

Range: $f(x) \in R$



SIMILAR TRIANGELS

Two triangles ABC and DEF are similar if they follow the following similarities.



1. **A-A-A :**
   If $\angle A = \angle D, \ \angle B = \angle E, \ \angle C = \angle F$
   then $\Delta ABC \sim \Delta DEF$

2. **S-S-S:**
   If $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$ then $\Delta ABC \sim \Delta DEF$

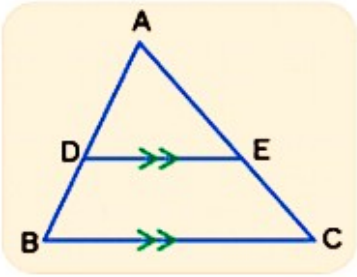
3. **S-A-S:**
   If $\frac{AB}{DE} = \frac{AC}{DF}$ and $\angle A = \angle D$ then $\Delta ABC \sim \Delta DEF$

Theorems related to Triangles :

Theorem-1 (Basic Proportionality theorem)

In a triangle line drawn parallel to one side divides the other two sides in the same ratio.

If DE || BC then $\frac{AD}{DB} = \frac{AE}{EC}$

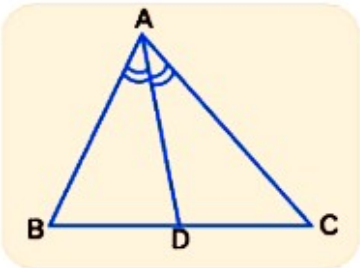


Theorem-2 (Angle bisector theorem)

The bisector of vertical angle of triangle divides the base in the ratio of other two sides.

If $\angle BAD = \angle CAD$

then $\frac{BD}{DC} = \frac{AB}{AC}$



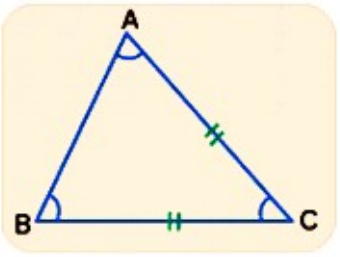
Theorem-3

Angle opposite to equal sides of a triangle are equal.

If AC = BC

then ∠B = ∠A

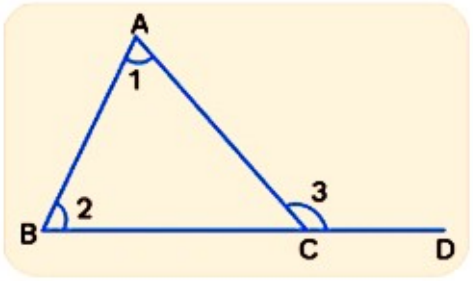
Also, **if ∠B = ∠A then AC = BC**



Theorem-4 (Exterior angle theorem)

If a side of a triangle is produced, the exterior angle so formed is equal to sum of the other two interior angles.

∠3 = ∠1 + ∠2

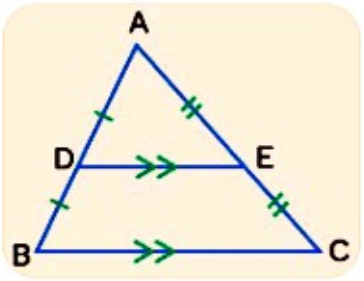


Theorem-5 (Mid-point theorem)

If the mid points of two sides of triangle are joined by a line segment, then this segment is parallel to the third side and is equal to half of it.

If AD = BD and AE = CE

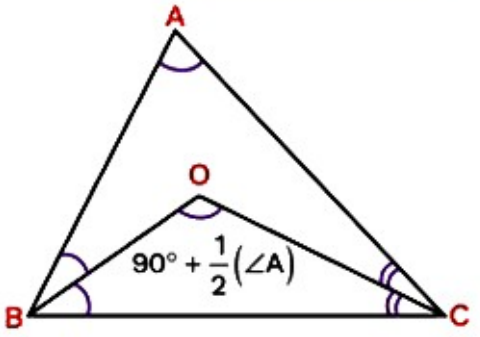
then DE || BC and DE = ½BC



Some useful results :

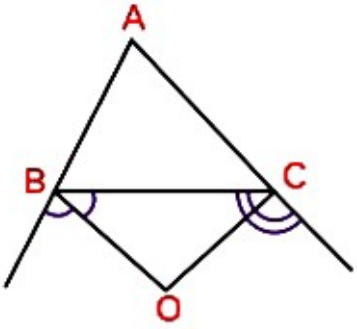
1. If in ΔABC interior angle bisectors of ∠B and ∠C meets at point O then

∠BOC = 90° + ½(∠A)



2. If in ΔABC exterior angle bisectors of ∠B and ∠C meets at point O then

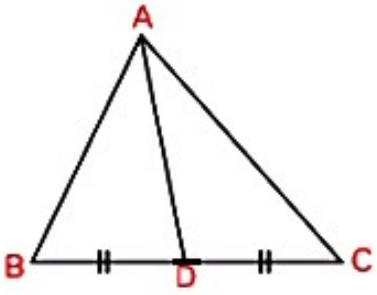
∠BOC = 90° − ½(∠A)



3. In ΔABC if AD is median then

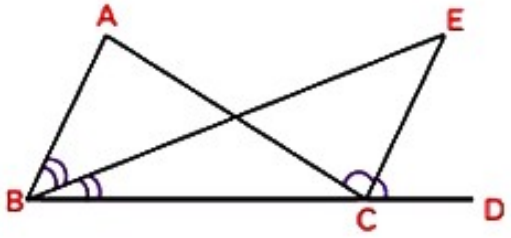
$AB^2 + AC^2 = 2(AD^2 + BD^2)$

(Apollonius's theorem)

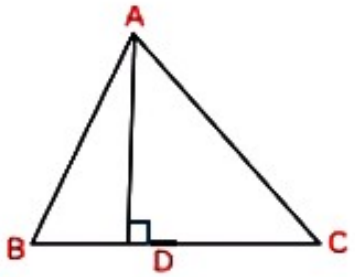


4. In ΔABC if BC is produced to D and bisector of ∠ABC and ∠ACD meets of E then

$\angle BEC = \frac{1}{2}(\angle BAC)$

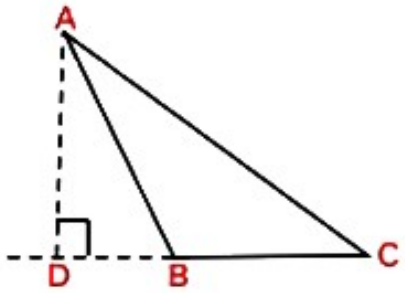


5. In an acute ΔABC if AD ⊥ BC then (B, D, C are collinear)



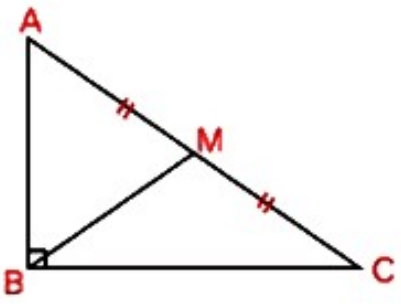
$AC^2 = AB^2 + BC^2 - 2BD.\ BC \quad (\angle B < 90°)$

$AC^2 = AB^2 + BC^2 + 2BD.\ BC \quad (\angle B > 90°)$



6. In ΔABC if ∠B = 90° and M is mid point of AC then

$BM = \frac{1}{2}(AC)$

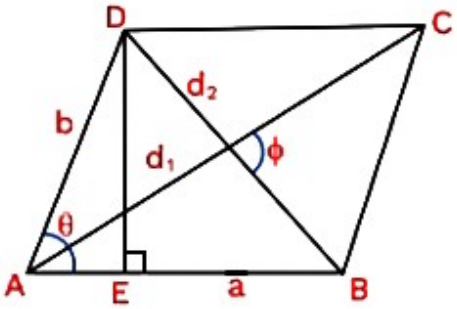


7. In ΔABC

(i) If $\angle B > \angle C$ then $b > c$
(ii) $\angle A + \angle B + \angle C = 180°$
(iii) $a + b > c$
(iv) $|a - b| < c$

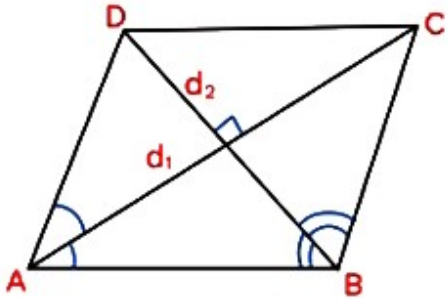
Quadrilaterals :

1. Parallelogram:



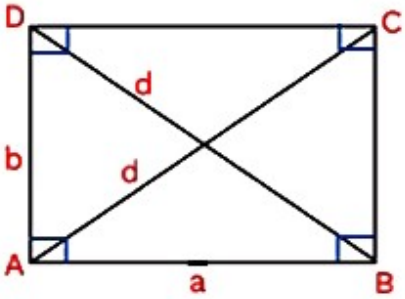
(i) Opposite side are parallel and equal.
(ii) Adjacent angles are supplementary.
(iii) Diagonals bisects each other.
(iv) Area $= ab\sin\theta = \frac{1}{2}d_1d_2\sin\phi$

2. Rhombus:



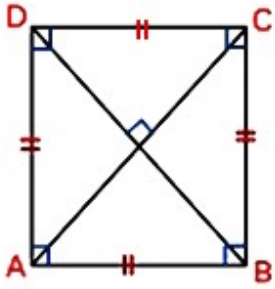
(i) All sides are equal.
(ii) Diagonals bisect at 90°.
(iii) Diagonals bisect angles.
(iv) Area $= \frac{1}{2}d_1d_2$

3. Rectangle:



(i) All internal angles are 90°.
(ii) Diagonals are equal.
(iii) $a^2 + b^2 = d^2$
(iv) It is a cyclic quadrilateral.

4. Square:

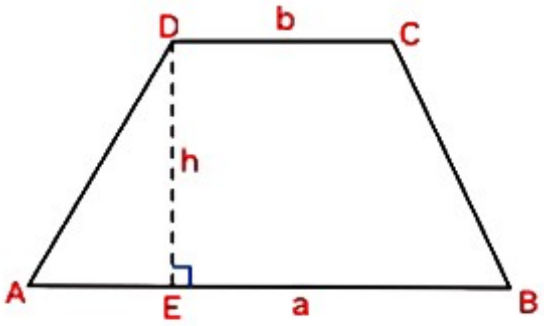


(i) All sides are equal.
(ii) Diagonals are equal and bisect each other at 90°.
(iii) Diagonals are angle bisectors.

5. Trapezium:

(i) One pair of opposite sides is parallel.

(ii) Area $= \frac{1}{2}(a+b)h$

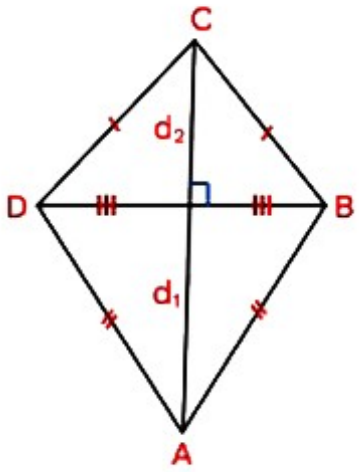


6. Kite:

(i) One diagonal divides the quadrilateral into two congruent triangles.

(ii) Diagonals are $\perp$.

(iii) Area $= \frac{1}{2} d_1 \cdot d_2$



7. Cyclic Quadrilateral:

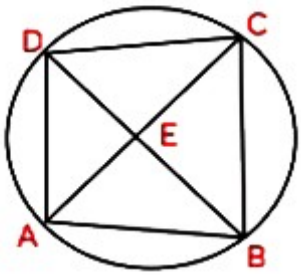
(i) Vertices lie on circle.

(ii) $\angle A + \angle C = \angle B + \angle D = 180°$

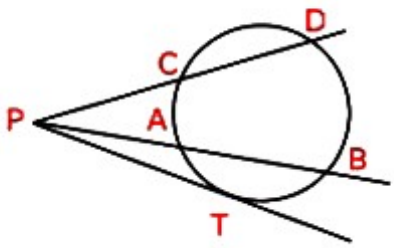
(iii) $AE \cdot EC = BE \cdot DE$

(iv) $AB \cdot CD + BC \cdot AD = AC \cdot BD$

(PTolemy's theorem)



8. $PA \cdot PB = PC \cdot PD = PT^2$



Circle:

(i) Equal chords subtend equal angles at centre and vice-versa.
(ii) Equal chords are equidistant from center and vice-versa.
(iii) Angle subtended by an arc at the center is double the angle subtended at any point on the remaining part of the circle.
(iv) Angles subtended by any chord on the same side on the circle are equal.
(v) The bigger chord will be closer to center.
(vi) A chord divides the circle into two segments.

Distance formula :

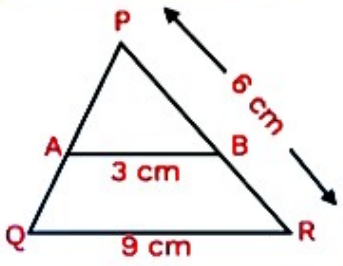
If $A(x_1, y_1)$ and $B(x_2, y_2)$ are two points in co-ordinate geometry, then distance between A and B

$$AB = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Note :

Sum of all interior angles of n-sided polygon is equal to $(n - 2)\pi$.

Q.1 In given figure AB || QR then find PB

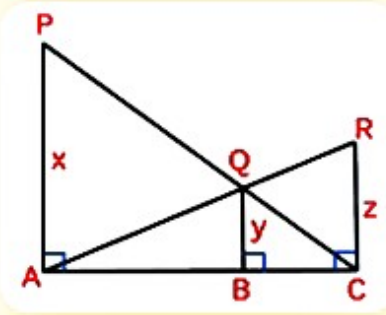


Sol. $\because$ AB || QR

$\therefore$ $\Delta PAB \sim \Delta PQR$

$$\frac{AB}{QR} = \frac{PB}{PR} \Rightarrow PB = \frac{3 \times 6}{9} cm = 2cm$$

Q.2 In the given figure, PA, QB, RC are perpendicular to AC then prove that $\frac{1}{x}+\frac{1}{z}=\frac{1}{y}$.



Sol. $\Delta PAC \sim \Delta QBC$

$$\frac{PA}{AC}=\frac{QB}{BC}\Rightarrow\frac{x}{AC}=\frac{y}{BC}\Rightarrow BC=\frac{y}{x}AC \quad ...(i)$$

$\Delta RCA \sim \Delta QBA$

$$\frac{RC}{AC}=\frac{QB}{AB}\Rightarrow\frac{z}{AC}=\frac{y}{AB}\Rightarrow AB=\frac{y}{z}AC \quad ...(ii)$$

$$(i)+(ii)\Rightarrow BC+AB=\left(\frac{y}{x}+\frac{y}{z}\right)AC$$

$$\Rightarrow\frac{1}{y}=\frac{1}{x}+\frac{1}{z}$$

Hence proved.

Q.3 If side of a rhombus is 10 cm and one of the diagonals is 16 cm, then find the length of other diagonal.

Sol. Let AB = 10 cm and AC = 16 cm $\Rightarrow$ OA = 8 cm

In ΔAOB

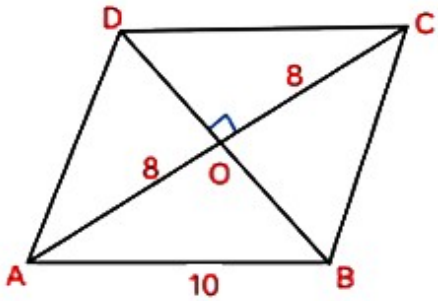
$\angle AOB = 90°$

$\therefore \quad OA^2 + OB^2 = AB^2$

$8^2 + OB^2 = 10^2$

$\Rightarrow$ OB = 6 cm

$\Rightarrow$ BD = **12 cm** (other diagonal)



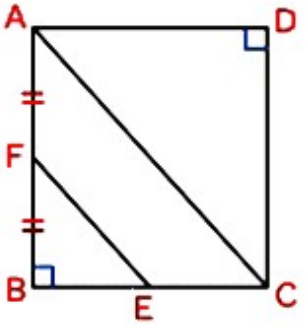
Q.4 ABCD is a square F is midpoint of AB. BE is one third of BC. If area of $\Delta FBE = 108\ cm^2$, then find the length of AC.

Sol. Let side of square AB = x cm

Now $FB=\frac{x}{2}cm, \quad BE=\frac{x}{3}cm$

$ar(\Delta FBE) = 108\ cm^2$

$$\frac{1}{2}BE\cdot FB=108\,cm^2$$



$$\frac{1}{2} \cdot x \cdot \frac{x}{2} = 108 \Rightarrow x = 36$$

Now $AC = \sqrt{AB^2 + BC^2} = \sqrt{x^2 + x^2} = \sqrt{2}x$

$AC = 36\sqrt{2}$ cm

Q.5 Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Sol. From the figure we know that

$OA = 5$ cm

$OB = 3$ cm

Now, in Δ OAB

$\angle OBA = 90°$

$OB^2 + BA^2 = OA^2$

$3^2 + BA^2 = 5^2 \Rightarrow BA = 4$ cm

Length of chord = $AC = 2AB = 8$ cm

